

Tu Yuang

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Robot Learning · Dexterous Manipulation · VLA Models · World Models · Reinforcement Learning

EDUCATION

Nanyang Technological University

Master of Science in Mechanical Engineering

Singapore

Aug. 2025 – Jan. 2027

Shanghai Jiao Tong University

Bachelor of Engineering in Energy and Power Engineering

Shanghai, China

Sep. 2021 – Jun. 2025

PUBLICATIONS

Y. Tu, R. Tan, Y. Yan, J. Hu, C. Lv. “AnyviewMeter: Adapting Robotic Reward Models with Camera Geometry and Multi-View Attention.” arXiv:2609.20106 *Submitted to ICRA 2027*

- Proposed a geometry-conditioned adaptation framework that keeps pretrained robotic reward models stable under camera viewpoint change and occlusion, supporting both single-view and joint multi-view reward prediction
- Injected camera geometry into visual features, queries, and keys via token-aligned Plücker ray conditioning, and fused synchronized views inside the decoder with parameter-free synchronous block attention plus a self-readout mask for cross-camera fusion
- Cut progress-prediction error by $\sim 21\%$ under a changed field of view versus RGB fine-tuning, and by 41–69% over averaged single-view prediction across simulated and real manipulation tasks

R. Tan, **Y. Tu**, Y. Yan, L. Yu, X. Tian, C. Lv. “FIERCE: From Generalist Robot Policies to Fast Specialists via Progress-Failure Feedback.” arXiv:2609.18651 *Submitted to ICRA 2027*

- Built a unified, task-adaptive progress-failure evaluator sharing an observation-language representation between an observed-progress head and an action-conditioned latent predictor feeding a causal failure-estimation head
- Derived progress shaping and failure-risk penalties from frozen evaluator snapshots, removing the need for continued generalist action queries, a target-task simulator, or hand-annotated dense rewards
- Validated on LIBERO and ManiSkill3 in simulation and on two contact-rich real-robot tasks (peg-in-hole, cup stacking), reaching 92% mean real-robot success vs. 45% for the original $\pi_{0.5}$, 80% for task-adapted $\pi_{0.5}$, and 83% for Robometer-guided RL

EXPERIENCE

Research Intern

Jun. 2026 – Present

*Advanced Remanufacturing and Technology Centre (ARTC), A*STAR*

Singapore

- Built a teleoperation data-collection platform (AlphaBot2 + exoskeleton) and the pipeline turning demonstrations into multi-task LeRobot datasets
- Fine-tuned $\pi_{0.5}$ on a multi-step PCB screw-fastening task for a wheeled bimanual humanoid, then froze it as the base policy producing base actions
- Trained a force-aware residual policy on top of the frozen $\pi_{0.5}$ for contact-rich hole alignment and fastening, taking visual features, proprioception, and force and torque signals as input and learning online on the real robot via SAC + RLPD with human interventions; a gate modulates the residual before composing it with the base action

Algorithm Intern

Jun. 2024 – Sep. 2024

Shanghai Ranjing Technology Co., Ltd.

Shanghai, China

- Trained deep learning models and evaluated human skeleton keypoint recognition performance
- Developed rule-based algorithms for multi-exercise repetition counting and violation detection
- Designed image-processing methods for standing long jump distance estimation

PROJECT EXPERIENCES

- Fine-grained Mobile Manipulation with VLA Models** | *Master's Thesis* Sep. 2025 – Present
- Targeted fine-grained tasks requiring coordinated base motion and bimanual manipulation; collected whole-body teleoperated multimodal demonstrations on MobileAloha and reproduced π_0 , ACT, and Diffusion Policy in LeRobot as baselines
 - Designed an MoT VLA with separate mobile and manipulation action experts, each with weights separate from the VLM backbone, modeling base and arm actions respectively
 - Proposed coupling-guided two-stage action generation: discrete flow matching first predicts a mobile-manipulation task-coupling level, which then sets the cross-expert attention bias for joint flow matching over base and arm actions
- AI for Industry Challenge by Intrinsic (Part of Google)** | *Core Member* Feb. 2026 – Aug. 2026
- Built an end-to-end HIL-SERL system for 6-DoF robot-arm two-end cable insertion on the AIC benchmark, implementing the full chain of demonstration collection → reward classifier → BC pretraining → RLPD online optimization → human-in-the-loop refinement, evaluated in simulation
 - Ranked 7th of ~120 teams in the Qualification Phase with a fully automated HIL-SERL workflow, achieving 95% two-end insertion success on five cables over repeated rollouts across varied rail poses
- RoboMaster Competition (World Champion)** | *Project Team Leader* Dec. 2021 – Aug. 2023
- Led the research, development, and maintenance of the infantry robot's mechanical structure (2022)
 - Led the quadrotor aerial robot project (2023): drone mechanical design in SolidWorks with lightweight, aerodynamic, and vibration optimizations based on simulation and testing
 - Captured image and IMU data with the Intel RealSense T265, configured sensor-fusion localization on the onboard computer, and tuned VIO/VINS-based visual positioning
- Flight Control of Kilogram Quadrotor Cyclocopter** | *Undergraduate FYP* Dec. 2024 – May 2025
- Developed a flight control algorithm for a quadrotor cyclocopter based on the PX4 platform, implementing PID control and a sensor-fusion EKF algorithm
 - Built a Gazebo plugin modeling the physical effects of the rolling wing and completed the full simulation model; achieved hovering and attitude control in SITL and HITL simulation
 - Designed the control-system hardware and electrical system of the aircraft and completed hardware wiring and debugging
- Modeling, Communication & Control of SCARA Robot** | *Research Program (PRP)* May 2022 – Sep. 2023
- Designed a machine-vision-based screw pre-tightening system for a SCARA robot, training a YOLOv8 model to detect screws and their types
 - Developed a robot end-effector capable of automatic bit switching with a force-feedback control algorithm to determine tightening levels
 - Performed forward/inverse kinematics analysis and completed robot modeling and simulation in MATLAB

TECHNICAL SKILLS

Programming & Software: Python, C++, MATLAB, Linux, ROS, Docker, Git
Robot Learning: PyTorch, JAX; imitation learning, reinforcement learning, VLA models
Simulation: Isaac Sim, OmniGibson, MuJoCo, ManiSkill, Gazebo
Engineering: End-to-end robot mechanical design and assembly, electrical system design and integration, real-robot deployment and debugging across multiple platforms

ACADEMIC AWARDS

2025 — NTU School of MAE Graduate Scholarship (~USD 40,000, top 1%)
2025 — China College Students' Mechanical Engineering Competition, National Third Prize
2024 — SJTU Academic Progress Scholarship
2023 — RoboMaster National University Robot Competition, National First Prize
2022 — RoboMaster National University Robot Competition, National First Prize
2022 — SJTU University-level Scholarship